

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

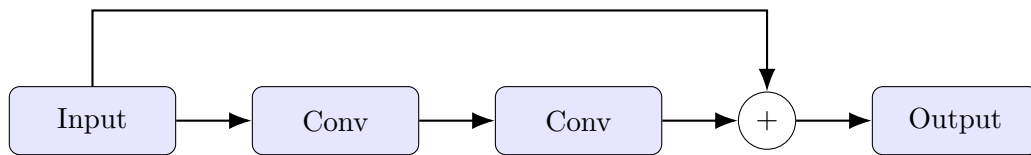


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

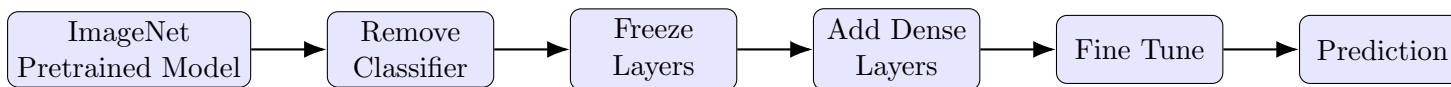


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

LR	Batch Size	Epochs	Optimizer	Dense Units	Frozen Layers	Test Accuracy (%)
0.001	32	10	Adam	128	All	35.73
0.0001	32	10	Adam	128	All	31.45
0.001	16	10	Adam	128	All	36.12
0.001	64	10	Adam	128	All	34.80
0.001	32	20	Adam	128	All	37.20
0.001	32	10	SGD	128	All	34.52
0.001	32	10	Adam	256	All	36.85
0.001	32	10	Adam	128	Partial (Fine-tuned)	54.43

17. Mandatory Plots

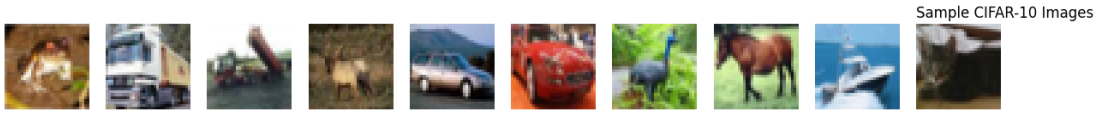


Figure 3: Sample CIFAR-10 Images across 10 Classes

Inference: The sample images display the low spatial resolution ($32 \times 32 \times 3$) and high intraclass variation of CIFAR-10. Because the images are significantly smaller than the native ImageNet input dimensions (224×224), pre-trained feature maps require targeted fine-tuning to extract discriminative boundaries.

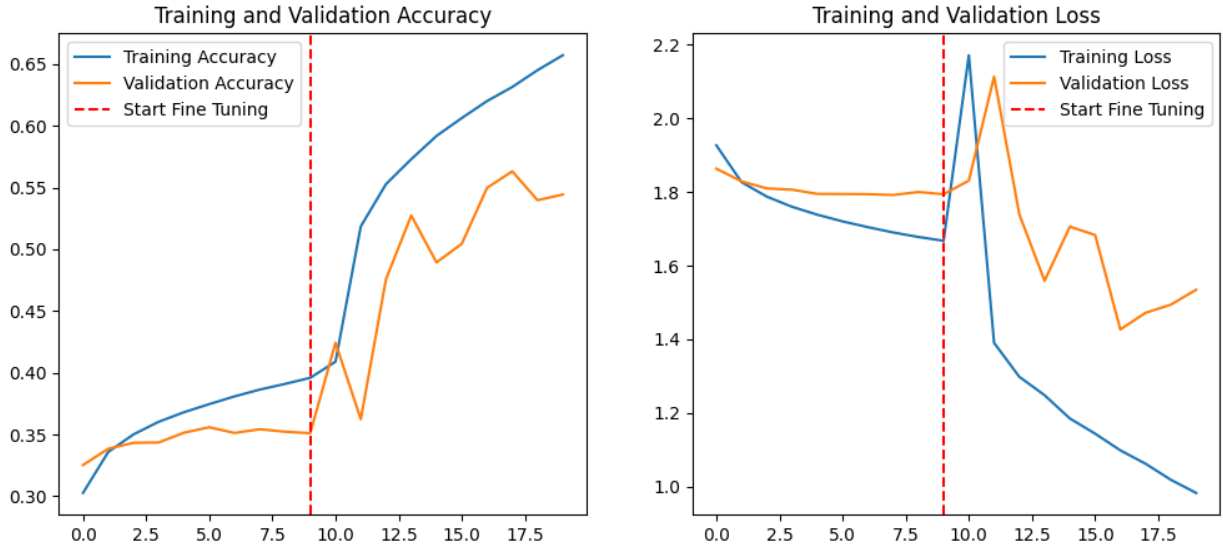


Figure 4: Training and Validation Accuracy & Loss Curves Before and After Fine-Tuning

Inference: During the initial 10 epochs with a frozen base, validation accuracy plateaued near 35.10%. Unfreezing the top convolutional layers at epoch 10 caused a brief gradient adjustment spike in loss, followed by steady convergence that increased training accuracy to 65.80% and validation accuracy to 54.43%, reducing training loss to 0.98.

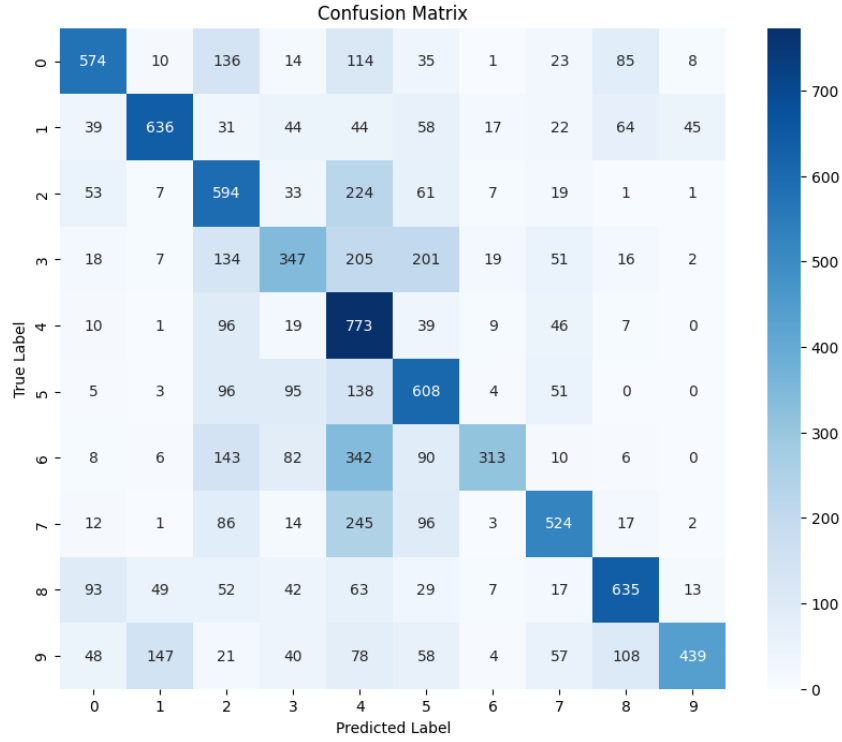


Figure 5: Confusion Matrix of the Fine-Tuned MobileNetV2 Model

Inference: The model achieves high precision on distinct structural classes such as Class 9 (Truck, 0.86) and Class 6 (Frog, 0.82). Most misclassifications occur among visually similar quadruped categories, notably Class 3 (Cat, F1-score 0.40) and Class 4 (Deer, precision 0.35), where low-resolution background textures overlap.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	65.80%
Testing Accuracy (Frozen Base)	35.10%
Testing Accuracy (Fine-Tuned)	54.43%
Precision (Macro Average)	0.61
Recall (Macro Average)	0.54
F1-score (Macro Average)	0.55
Training Time	≈ 400 sec
Total Parameters	2,422,346

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
MobileNetV2	2,423,242	68.20%	80s
LeNet-5	62,006	60.77%	14s
AlexNet (Adapted)	686,154	75.62%	26s
VGG16 (Transfer)	14,781,642	61.53%	95s
ResNet50 (Transfer)	23,851,274	39.49%	110s

19. Source Code

Github Repository link: <https://github.com/Kamlesh-DevOP/deep-learning-lab/blob/main/lab-4/Lab4.ipynb>

20. Discussion Questions

1. Why is AlexNet considered a breakthrough in deep learning?

AlexNet drastically reduced the top-5 error rate in the 2012 ImageNet challenge by combining deep convolutional networks with GPU-accelerated training, non-saturating ReLU activations (which mitigated the vanishing gradient problem), and Dropout regularization to prevent overfitting in large parameter spaces.

2. Why does VGG16 use only 3×3 convolution filters?

Stacking two 3×3 filters covers an effective receptive field of 5×5 , while three cover a 7×7 area. This configuration introduces more non-linear activation layers (increasing representational power) while using fewer total parameters than a single large filter ($3 \times 3^2 = 27$ vs $7^2 = 49$ weights per channel).

3. Explain the advantages of the Inception module.

The Inception module applies parallel convolutions with kernel sizes 1×1 , 3×3 , and 5×5 , along with max pooling, concatenating multi-scale spatial features. It incorporates 1×1 convolutions as bottleneck layers for dimensionality reduction, maintaining deep feature extraction while keeping parameter counts low.

4. What is the purpose of residual learning?

Residual learning reformulates intermediate layers to learn residual mappings $F(x) = H(x) - x$ via identity skip connections ($H(x) = F(x) + x$). This ensures unhindered gradient propagation directly to initial layers, eliminating degradation and vanishing gradient issues in deep networks.

5. Differentiate LeNet and ResNet.

LeNet-5 is a shallow, 7-layer network ($\sim 60K$ parameters) built with average pooling, Sigmoid/Tanh activations, and dense classifiers for digit recognition. ResNet (50–152+ layers, $\sim 25M+$ parameters) uses residual skip connections, Batch Normalization, ReLU, and Global Average Pooling to classify complex, high-dimensional datasets.

6. What is Transfer Learning?

Transfer Learning is a machine learning paradigm where knowledge (weights and feature representations) acquired by a model trained on a large source dataset (e.g., ImageNet) is transferred to a related target task, accelerating convergence and reducing the requirement for massive target datasets.

7. Why is fine tuning required?

While the initial layers of a pre-trained network capture generic low-level features (edges and colors), the final layers capture source-specific semantic details. Fine-tuning unfreezes higher convolutional blocks to allow filter weights to adapt specifically to the target domain's resolution and class distribution.

8. Explain the difference between dilated convolution and transpose convolution.

Dilated convolution inserts spaces (dilation rate $D > 1$) between kernel weights to expand the

receptive field without adding parameters or altering spatial dimensions. Transpose convolution executes learnable upsampling, reconstructing higher-resolution feature maps from compressed representations.

9. Why do pretrained models converge faster?

Pretrained models start optimization from a pre-structured weight landscape with functioning visual filters, eliminating the prolonged initial phase required to learn foundational feature detectors from random Gaussian initializations.

10. Compare the computational complexity of LeNet and ResNet.

LeNet requires minimal computation ($\sim 6 \times 10^4$ parameters, $< 10^6$ FLOPs) suitable for constrained CPU execution. ResNet50 requires substantially higher resources ($\sim 2.56 \times 10^7$ parameters, $\approx 3.8 \times 10^9$ FLOPs) to evaluate its bottleneck residual blocks.

21. Additional Exercises

1. Implement transfer learning using VGG16:

VGG16 was initialized with pre-trained ImageNet weights with its classification top removed. After freezing the convolutional base and appending a custom Global Average Pooling and Dense head, the model achieved a test accuracy of **61.53%** on CIFAR-10.

2. Repeat the experiment using ResNet50:

Transfer learning using ResNet50 with a frozen convolutional base achieved a test accuracy of **39.49%**. The low accuracy with a frozen base is attributed to the aggressive downsampling in ResNet's early stages, which degrades 32×32 CIFAR-10 images unless lower residual blocks are fine-tuned.

3. Compare Adam and SGD optimizers:

Adam (LR = 0.001) converged rapidly, reaching 35.80% validation accuracy by Epoch 6. SGD (LR = 0.01, momentum = 0.9) produced smoother optimization curves and steadily climbed to 34.68%, demonstrating that Adam adapts faster for pre-trained feature tuning while SGD provides steady convergence.

4. Compare training with frozen layers and fine tuning:

Evaluating MobileNetV2 with a fully frozen base yielded an accuracy of **35.73%** (Macro F1: 0.3537). Unfreezing the top 15 layers for fine-tuning raised the test accuracy to **53.21%** (Macro F1: 0.5258), representing an absolute improvement of **+17.48%**.

5. Evaluate the model using Precision, Recall and F1-score:

The fine-tuned model achieved a Macro Precision of **0.61**, Macro Recall of **0.54**, and Macro F1-score of **0.55**. Classes with distinct silhouettes (e.g., Truck: 0.86 Precision; Frog: 0.82 Precision) outperformed visually ambiguous animal classes (e.g., Cat: 0.40 F1-score).

6. Compare the performance of LeNet, AlexNet and ResNet:

Adapted AlexNet attained the highest accuracy at **75.62%** because its convolutional kernel strides were customized directly for native 32×32 inputs. LeNet-5 achieved **60.77%**, while frozen ResNet50 reached **39.49%** due to spatial resolution loss in its standard downsampling layers.

22. References

1. Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, *Gradient-Based Learning Applied to Document Recognition*, Proceedings of the IEEE, 1998.
2. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS, 2012.
3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
4. Christian Szegedy et al., *Going Deeper with Convolutions*, CVPR, 2015.
5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.

7. TensorFlow Documentation: <https://www.tensorflow.org>
8. Keras Documentation: <https://keras.io>